



SECR1213-12 (Network Communications)

Task 4: Making The Connections

Group 5

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Date of Submission: 28/1/2025

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INTRODUCTION

The design and implementation of a reliable network infrastructure are essential for ensuring seamless connectivity, across the faculty. This task focuses on planning and deploying the appropriate devices, cables, and configurations to support the diverse needs of the building's work areas. By integrating Both wired and wireless networks, the setup aims to provide robust performance and scalability for academic and administrative functions.

The network design considers the specific requirements of each work area, including labs, classrooms, and recreational spaces. High speed Ethernet cables and fiber optic backbones have been selected to ensure efficient communication across floors and devices. Wireless access points are strategically placed to complement the wired network, offering flexibility for mobile users. This comprehensive approach ensures a reliable and future-proof network infrastructure for the faculty.

1.0 Work Area on the floor plan

1.1 Ground Floor

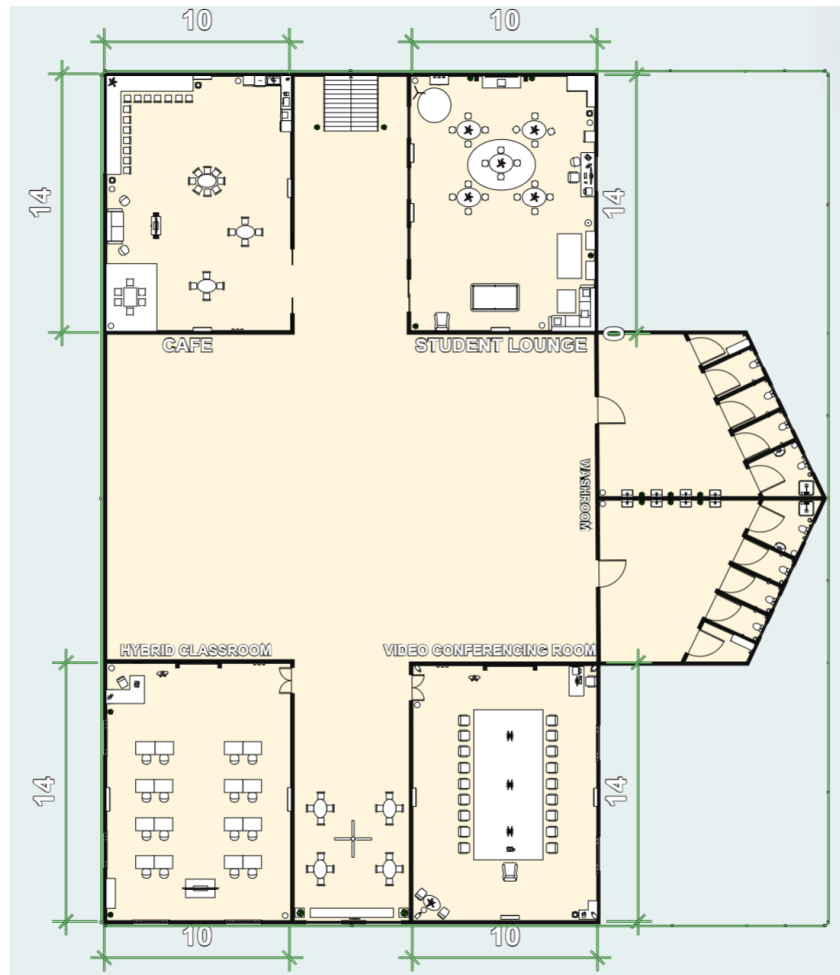


Figure 1.1 Floor plan of ground floor

The figure shown above is the ground floor plan of the new building of the Faculty of Computing which consists of several work areas including Hybrid Classroom, Video Conferencing Room, Cafe and Student Lounge. These work areas have their own specific function for students and staff in the building. Hybrid Classroom will be used for hybrid mode teaching and learning. This room is equipped with necessary teaching devices to cater for hybrid mode. There is also a Video Conferencing Room specially designed for conference and also a formal meeting. High-end devices are provided to ensure a smooth and efficient operation. Additionally, there is a student lounge that offers a comprehensive internet access designed for students to relax and group discussion as well as a cafe for students and staff to take a break and have a quick drink during their leisure time.

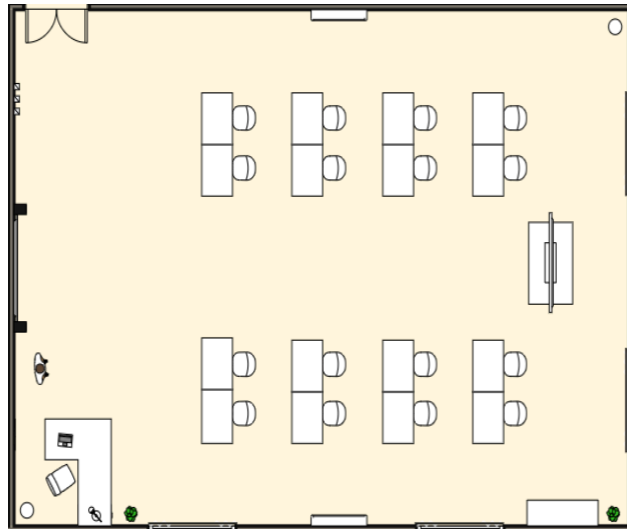


Figure 1.1.1 Hybrid Classroom

For Hybrid Classroom, it is designed for hybrid mode of lecture sessions for students to enhance the learning experience. This room can be occupied by up to 16 students and two lecturers maximum. The devices provided in this room are two Smart TVs. One for hosting an online meet and another one to be used as a digital whiteboard and sharing screen in the online meeting. The Smart TVs will be connected to one patch panel and one switch and they will be connected to wireless access point. Thus, the room will have a good WiFi connection and ensure that the learning process runs smoothly without any interruption or laggy connection.

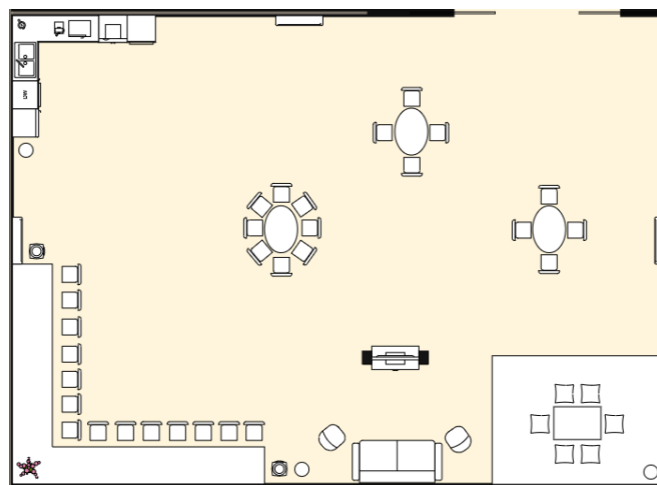


Figure 1.1.2 Cafe

Cafe serves as a place for both students and staff to enjoy their break time and have a chit chat with their friends while grabbing some drinks. This particular space does not really have any technology integrated devices but there is a Smart TV for students and staff to watch their favourite shows. This room will have a patch panel, a switch and also a wireless_ access point to provide a seamless WiFi connection for their visitors to ensure their favourite shows run very well.

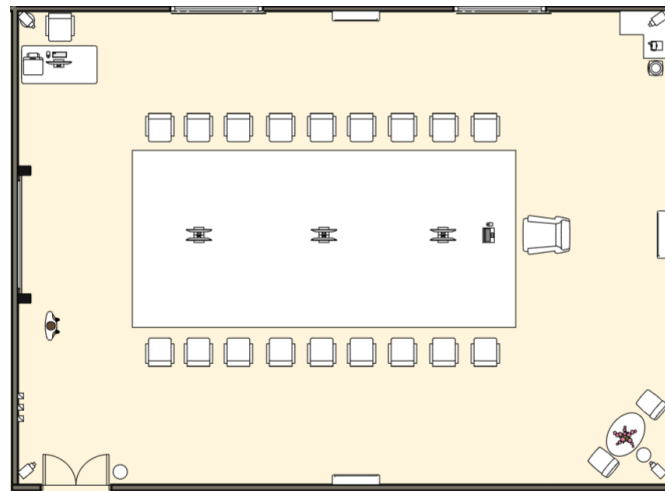


Figure 1.1.3 Video Conferencing Room

This video conferencing room is designed to facilitate virtual meetings and collaborative sessions for students, faculty, and external participants. It can accommodate up to 19 people seated around a central conference table, one person on the main desktop to monitor and two people on the back of the room. The room is equipped with a Smart TV and high-definition video conferencing system to ensure smooth communication during virtual meetings. This room also comes with 7 monitors on the central conference table for participants to see the display on the Smart TV. This room also provides a smaller seating arrangement in the corners for private calls or discussions. This room is connected to a patch panel and a network switch. A wireless access point is also strategically placed to provide uninterrupted internet connectivity.

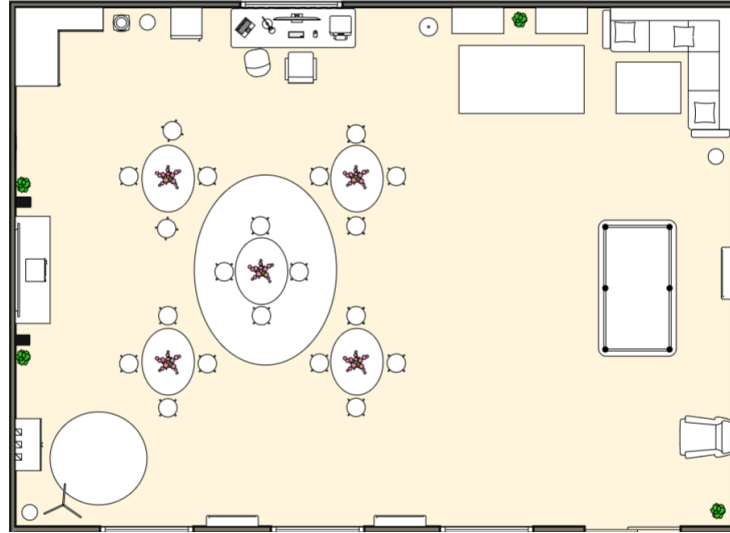


Figure 1.1.4 Student Lounge

The student lounge serves as a relaxation and socialization area for students. It is designed with multiple seating options, including sofas and chairs. A large Smart TV is provided and a pool table, enhancing the recreational aspect of this space. The lounge is equipped with a patch panel and a network switch as well as a wireless access point to provide seamless WiFi coverage. The connectivity setup guarantees a smooth internet experience, allowing students to relax or work without interruptions.

1.2 First Floor

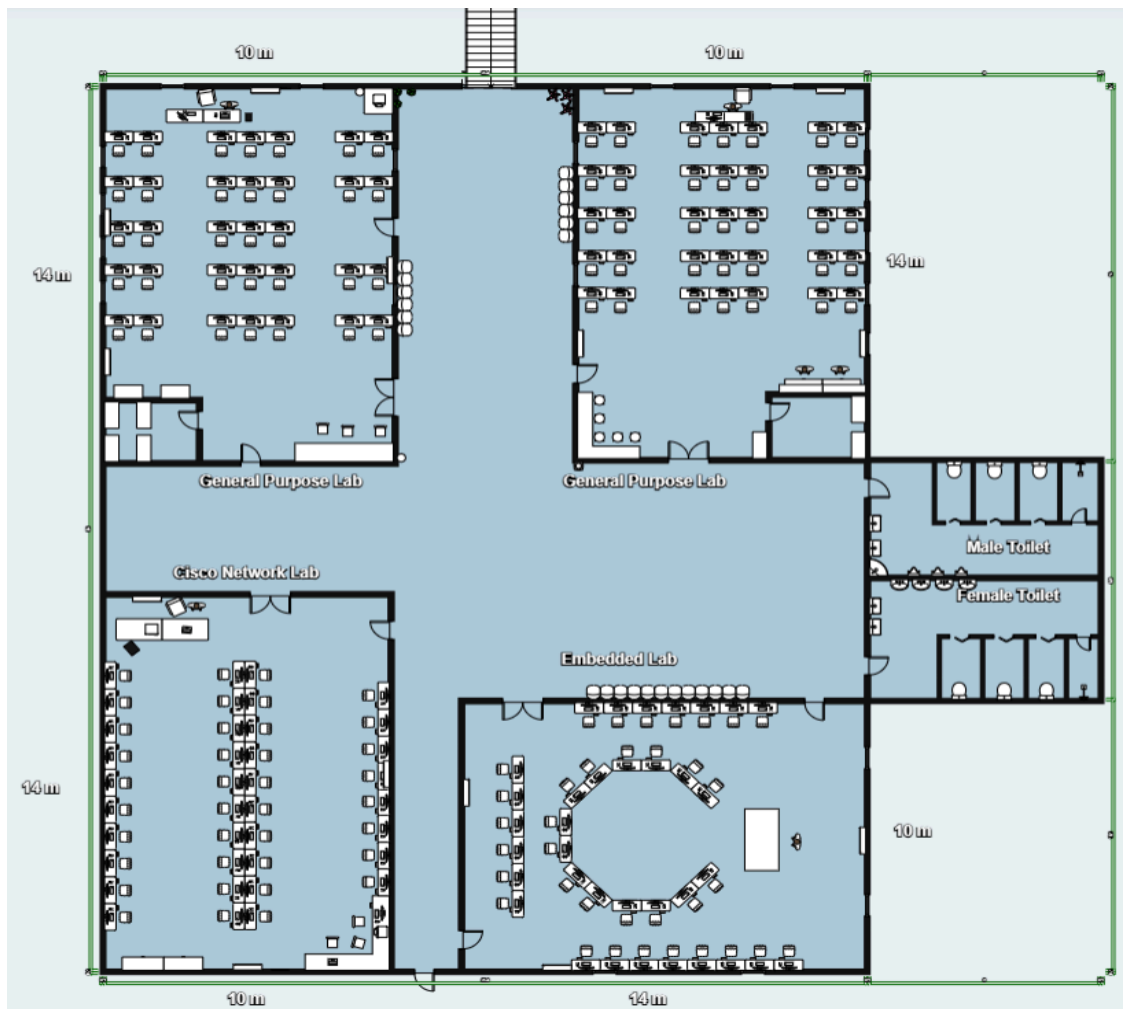


Figure 1.2 Floor plan of first floor

It includes a number of specialized labs and much-needed amenities in the first floor, where activities are smoothly carried out in order to achieve the ultimate academic goals by both students and faculty. The first floor is designed to house General Purpose Labs 1 and 2, the Cisco Network Lab, and the Embedded Lab—all specifically fitted for general computing, networking, and embedded systems programming. Each lab is well-equipped with the latest technologies, including servers, switches, patch panels, and wireless access points. All are placed to guarantee effective, high-speed connectivity. The design makes moving from lab to lab a smooth process that minimizes friction between sessions for the students. There are male and female restrooms very close to the labs for increased accessibility of the space. The design also distributes power and data to provide lines dedicated to computing for every lab's intensive needs. The central design and proximity to resources allow for sharing, thereby fostering collaboration in making the floor a hub of academic and technical excellence.

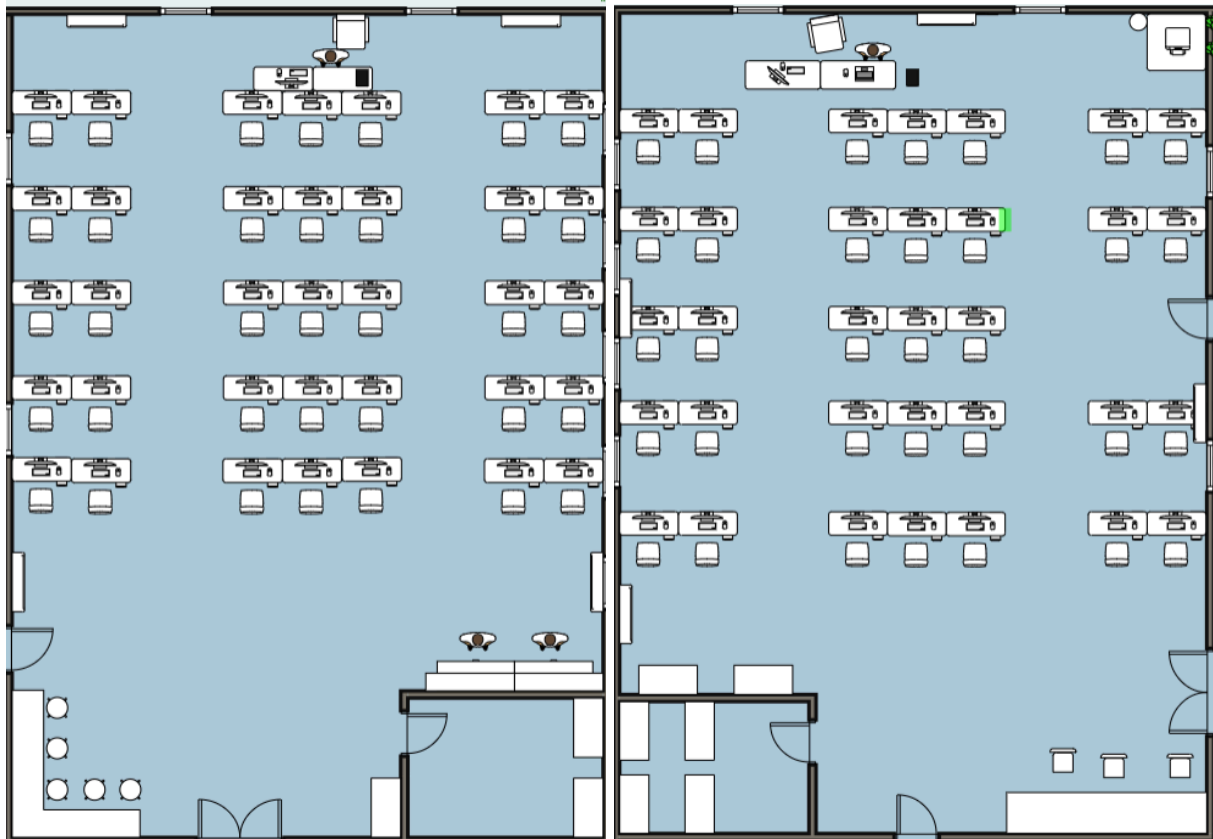


Figure 1.2.1 General Purpose Lab 1 & 2

General Purpose Lab 1 and 2 combine in a flexible layout where everything from academic to technical activities are effectively carried out. In each lab, 30 workstations are accommodated in structured rows for focused working by the students. The labs house servers, switches, patch panels, and access points for wireless connections of all devices. Classroom projectors allow the instructor to show presentations and other instructional materials, including live demonstrations of complex concepts. Labs are configured for both individual work and team projects, providing support for activities including but not limited to programming, software development, data analysis, and general computing. The flexibility of the physical setup allows smooth transitions between the various modes, flowing directly between individual work to group discussions and vice versa.

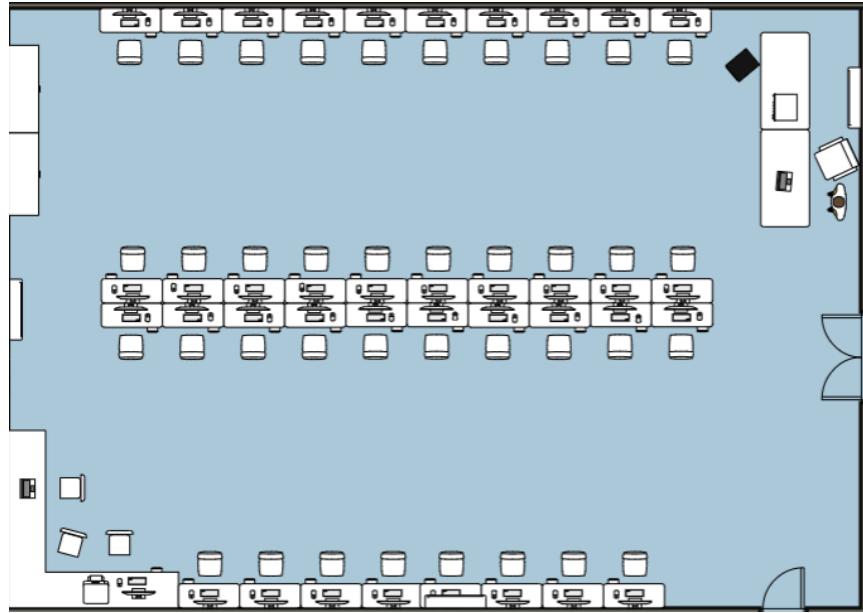


Figure 1.2.2 Cisco Network Lab

The Cisco Network Lab is a specially designed facility that has been equipped to offer hands-on training in network configuration, management, and troubleshooting. It contains 40 workstations in a structured layout, offering spacious and efficient working for both individual learning and collaborative projects. The lab houses advanced Cisco networking devices: routers, switches, and network simulation tools—one of the essential means for practicing real-world networking setups. The infrastructure contains a central server, patch panel, and switch for smooth data transmission, while there is a wireless access point to distribute high-speed internet all over the lab. Besides that, the projector enhances instruction by allowing tutors to provide step-by-step demonstration and real-time troubleshooting guidance. This lab is ideal for students pursuing certifications like CCNA and CCNP, as it allows them to work on advanced network designs, security protocols, and routing configurations. The layout encourages structured learning while accommodating collaborative group tasks, making it a hub for advanced networking education and practical exercises.

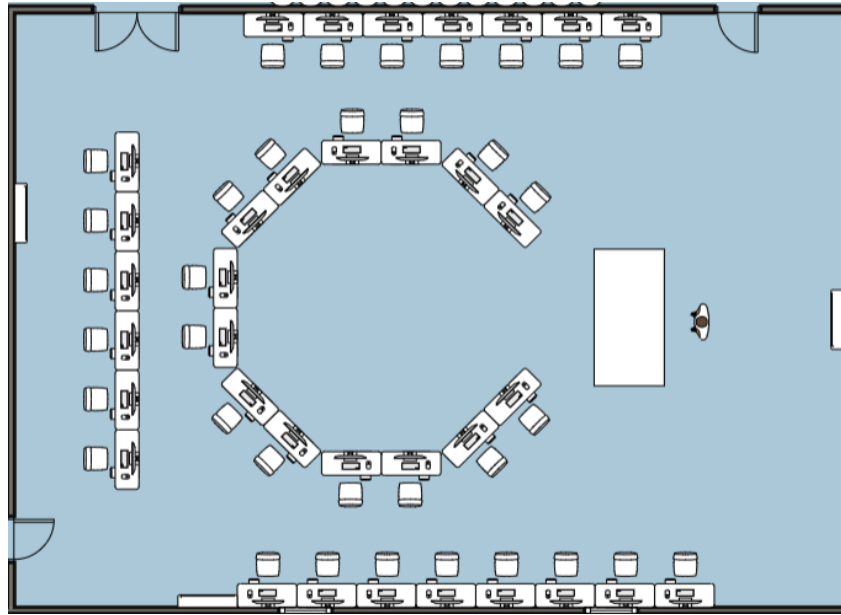


Figure 1.2.3 Embedded Lab

The Embedded Lab is a state-of-the-art facility that accommodates students in practically engaging with embedded systems and hardware programming. It houses up to 32 students, who will be provided with workstations that are strategically positioned in circular and linear formations to facilitate collaborative work while having dedicated workspaces at the same time. The lab also provides cutting-edge devices, including a central server, switch, patch panel, and wireless access points for flawless and faultless connectivity. There is also a central projector available that could enable live demonstration-the instructor should actually be showing in real-time, coding, circuit designs, or system integrations with ease to see for all the students. The inner workstations, set out in a circular format, are better for teamwork and active discussion, while the outer ones provide enough space for focused individual work. This lab will be useful in a variety of activities, including microcontroller programming, circuit design, robotics development, and Internet of Things projects. Equipped with reliable connectivity and an ergonomic layout, this provides the best learning environment for project-based tasks, ensuring that students can work without distractions or technical disruptions.

2.0 Network Diagram

2.1 Ground Floor

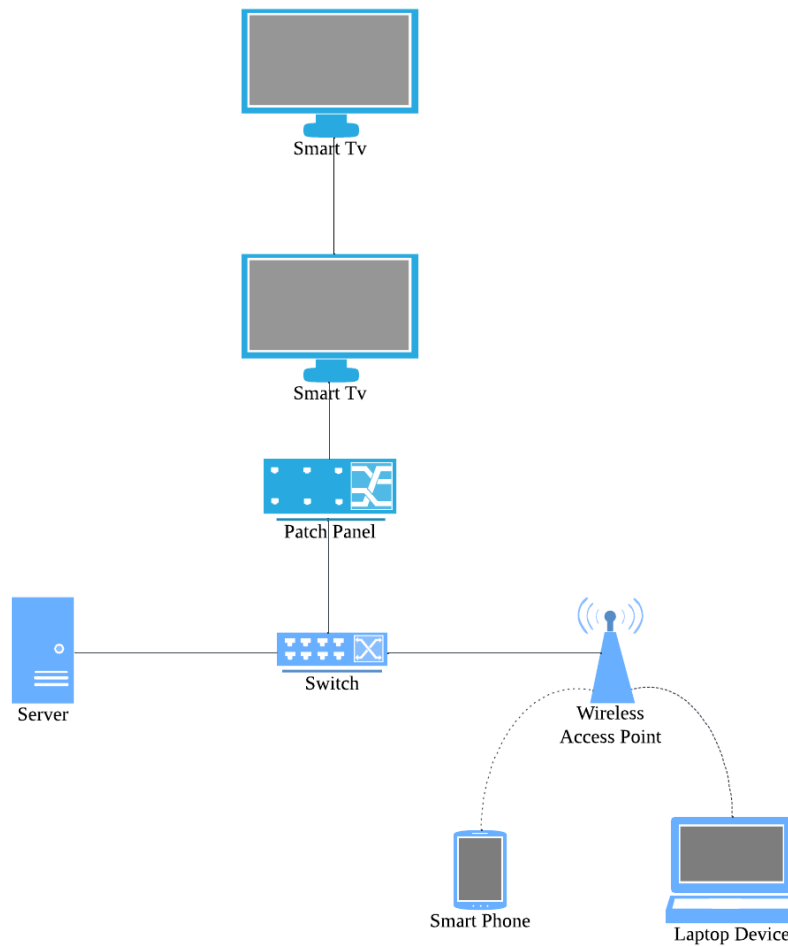


Figure 2.1.1 Network Diagram in Hybrid Classroom

This Hybrid Classroom setup consists of two smart TVs which play a crucial role to ensure the learning process runs effortlessly without any interference. Both of the TVs are connected to the server and Wireless Access Point through the patch panel and switch. Type of cable that is being used in this room is CAT6 cable, represented by the black line to avoid any signal or connection issues. Wireless Access Point also allows wireless connectivity for both students and lecturers to use wireless devices such as mobile phones or laptops. The server is responsible for providing services and resources such as files resources, applications or any other services needed for lecturing sessions. Students and lecturers can access the resources and information from the server which acts as a centralized storage. The switch is used to

connect multiple devices within the same network to facilitate communication between them. For instance, smart TVs and servers are both connected to the switch. The switch will then forward data from the TVs to the server if it is necessary or connect the TV to Wireless Access Point. Additionally, cables from all the devices in the room will go into the patch panel first before going into the switch, which will centralize the cable management and for easier maintenance if there are any additions or changes in the network.

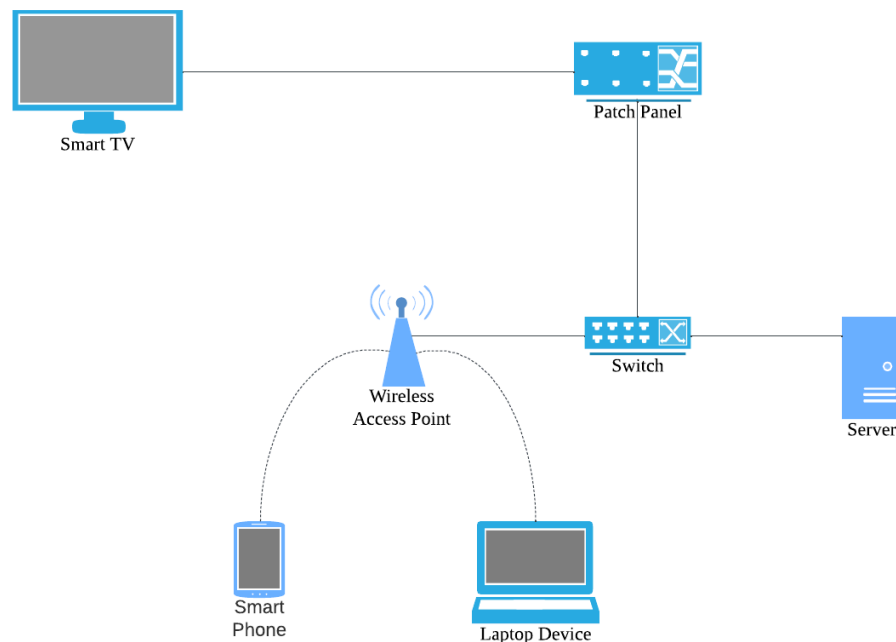


Figure 2.1.2 Network Diagram in Cafe

For the cafe, it has a smart TV for students or lecturers to use while they visit this place. The smart TV is connected to the Wireless Access Point through the patch panel and switch. Wireless Access Point provides a wireless connection for both smart TV and wireless devices used by the visitors. The server provides services mainly for applications as the TV is particularly used for watching shows from Netflix or any other apps. On the other hand, the switch is used to connect the smart TV with the server as well as Wireless Access Point. This will ensure network efficiency and seamless operation within the network. Lastly, the cables from smart TV, switch, Wireless Access Point and server are terminated into the patch panel. This will simplify the cable management and maintenance of the network. Additionally, the type of cable used in this network are also CAT6 cable to provide the best connection to all the devices.

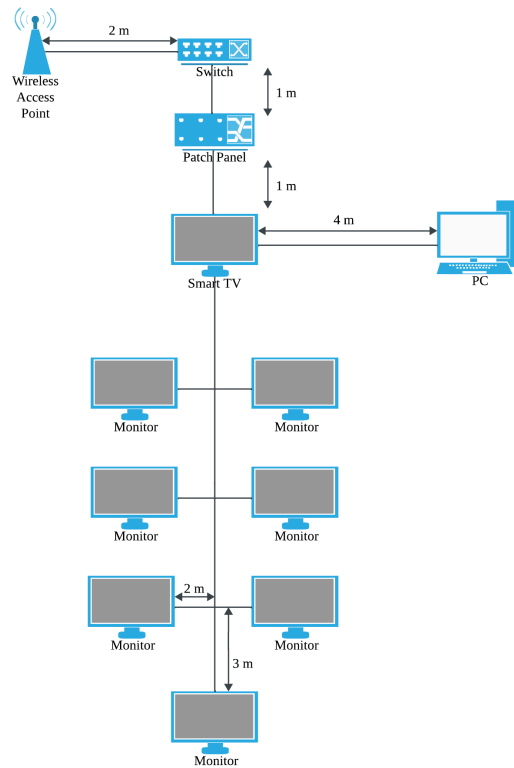


Figure 2.1.3 Network Diagram in Video Conferencing Room

The video conferencing room is equipped with a switch connected to a patch panel, a Smart TV desktop PC, and monitors to support seamless virtual meetings. The Smart TV serves as the primary display for video calls, while the desktop PC manages conferencing software. Each monitor is connected to the Smart TV to mirror its display, ensuring that all participants have a clear view of the content regardless of their seating position. A wireless access point ensures reliable WiFi for mobile devices, while Cat6 cables provide high speed wired connections. This setup guarantees stable and efficient connectivity, making the room ideal for professional and academic conferencing.

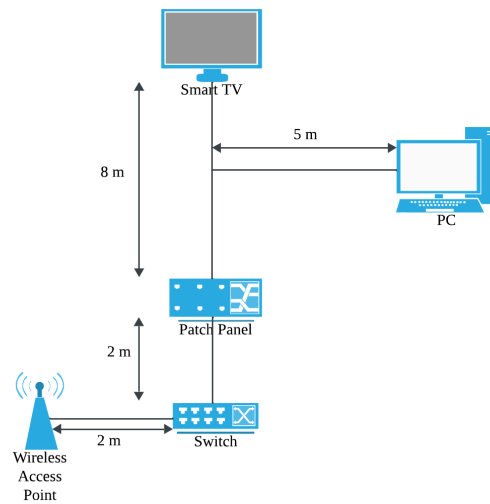


Figure 2.1.4 Network Diagram in Student Lounge

The student lounge is equipped with a Smart TV and a desktop PC each serving independent purposes. The Smart TV is used for entertainment, such as streaming video or displaying announcements, while the desktop PC is available for general use, such as browsing, completing assignments, or accessing shared resources. Both devices are connected to a patch panel, which is linked to a central switch for efficient wired network management. A wireless access point (WAP) is also installed to provide seamless Wi-Fi coverage for students personal devices, such as laptops or smartphones. The setup ensures both reliable wired and wireless connectivity, creating a space for relaxation and productivity.

2.2 First Floor

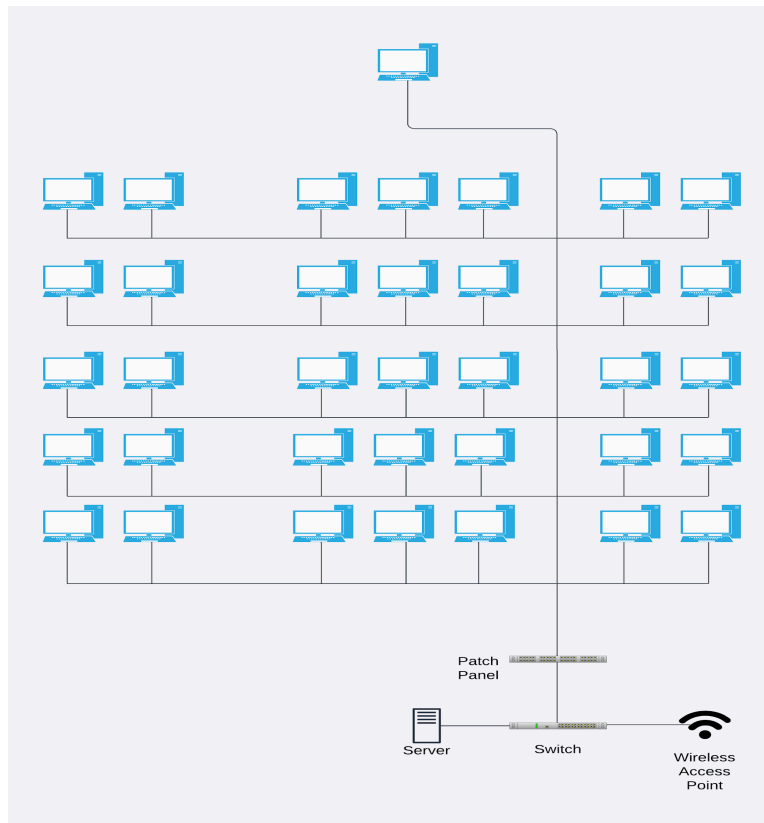


Figure 2.2.1 Network Diagram in General Purpose Lab 1

General Purpose Lab 1 is a multi-purpose lab that is meant for supporting different academic activities: programming, software development, and general computing. The lab configuration should feature 30 workstations connected using CAT6A Ethernet cables to ensure high speeds and stability of the internet. A length of 74 meters in total will be sufficient to provide stable Internet to this lab even during extreme use. The workstations are networked through a patch panel and switch ports, thus allowing smooth information flow and control of the network from one point. Further, this is complemented with a wireless access point that will let students and instructors connect their devices wirelessly without any compromise on speed or reliability. A projector is included to allow for interactive teaching, group discussions, and live demonstrations of programming or software applications. The ergonomic design of the lab allows for focused learning and yet allows collaboration on group projects. The infrastructure supports a variety of educational and technical tasks, such as simulations, research, and cloud-based computing. Furthermore, the setup is optimized for minimum network latency to make sure students can work efficiently on real-time applications and large data sets.

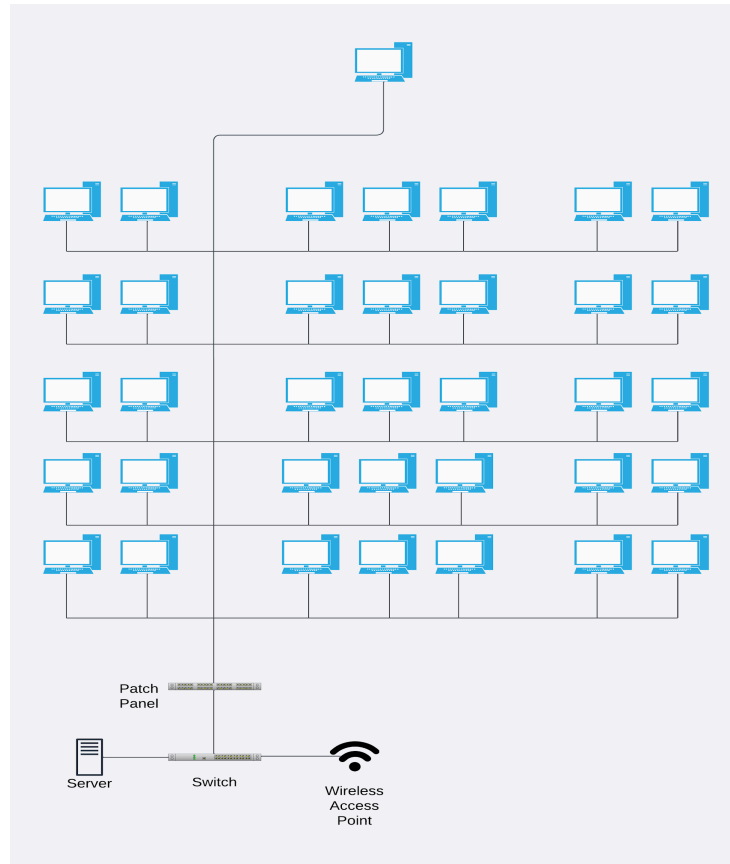


Figure 2.2.2 Network Diagram in General Purpose Lab 2

General Purpose Lab 2 has a similar design to General Purpose Lab 1. This lab also provides a general-purpose environment to support individual and collaborative academic activities. It has 30 workstations interconnected with CAT6A Ethernet cables spanning a length of 74 meters, ensuring fast data transfer rates among the computers connected over the network. The network is managed via a patch panel and switch ports to ensure smooth flow and centralized network control. A wireless access point ensures reliable access to the internet for students and instructors, thus enabling full utilization of online tools and resources. The teaching experience is immensely enhanced with a projector that allows instructors to present visual materials, show live demonstrations, and hold interactive sessions. The lab infrastructure is quite appropriate for programming assignments, software training, and collaborative group work; there is enough space for students to work comfortably. With its powerful connectivity and an efficiently laid-out facility, the students will have the potential of working on anything from cloud-based software development, data analysis, collaborating remotely with the other students and researchers, to tasks that require more computational resources.

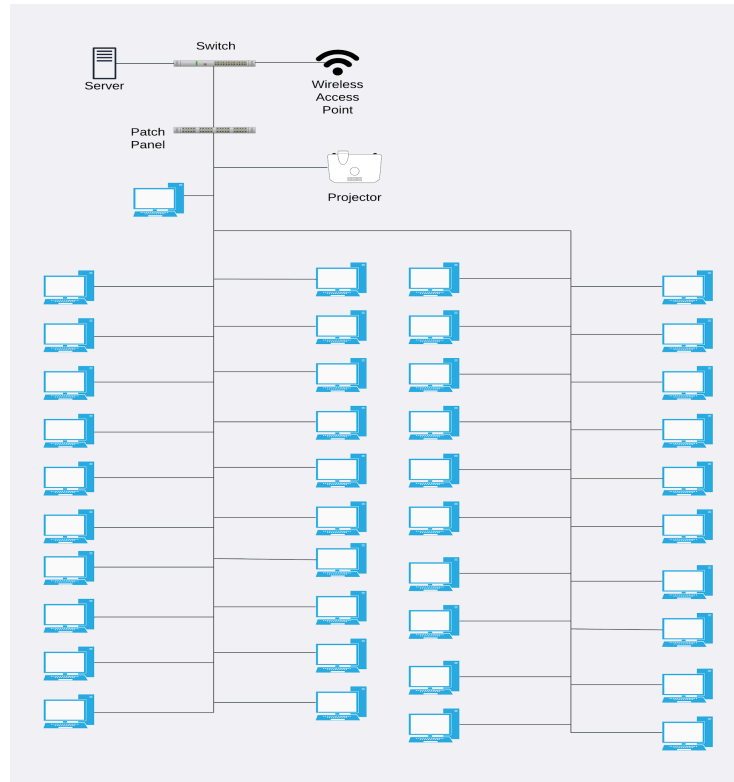


Figure 2.2.3 Network Diagram in Cisco Network Lab

The Cisco Networking Lab is a specially set-up facility for hands-on experience in gaining practical knowledge of networking concepts, configurations, and troubleshooting. This lab provides an optimized environment with 40 workstations connected by 46 meters of CAT6A Ethernet cables for practical learning. The network infrastructure consists of a patch panel and switch ports to manage the network reliably and scalably. A wireless access point ensures smooth and constant access to the internet, which students can use to access online resources, network simulation tools, and collaboration platforms.

The lab is further supported with sophisticated Cisco routers and switches that provide hands-on experience on how to configure networks, design topologies, and set up securities. It contains a projector at the front of the room where instructors can visually walk through demonstrations, show troubleshooting sessions, and demonstrate real-time network simulations. The structured layout ensures each student has a workspace that is also inviting for teamwork on group projects. It is an ideal lab for students interested in industry-reputed certifications such as CCNA and CCNP, since one will be practicing networking in a more realistic setting. The low-latency infrastructure greatly enables uninterrupted testing of the scenarios that might happen in a real-world network environment.

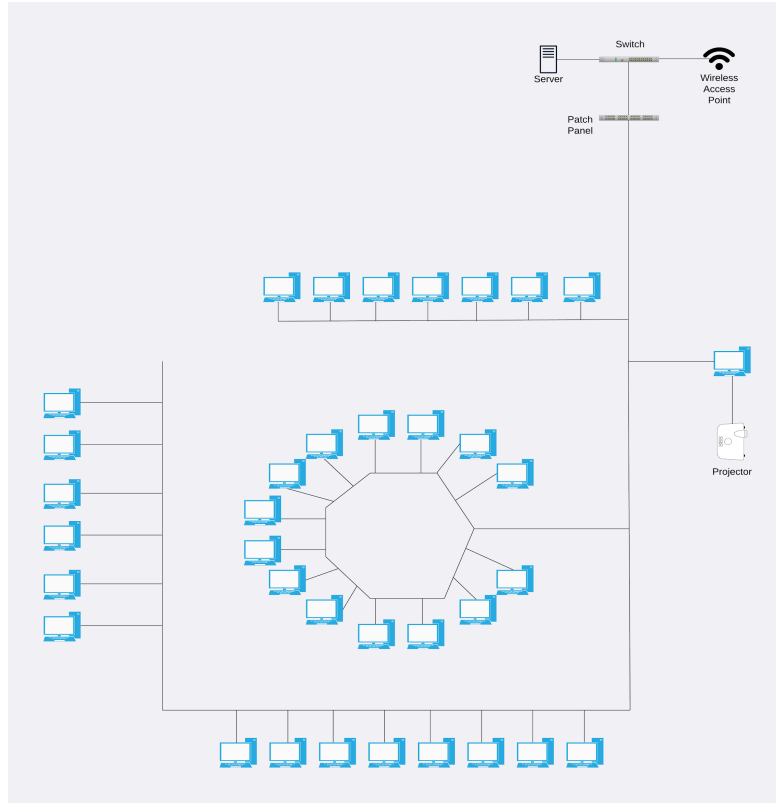


Figure 2.2.4 Network Diagram in Embedded Lab

The Embedded Lab is targeted for students and professionals working in the domain of embedded systems, robotics, and hardware programming. This lab features 32 workstations, arranged in both a central octagonal configuration and additional rows along the perimeter. In addition, this layout allows both team collaboration as well as focused individual work to take place in comfort. The lab interconnects with 59 meters of CAT6A Ethernet cabling, providing high-speed connectivity across all devices. That will be enabled by use of the network supported with patch panel, switch ports, and wireless access point for seamless connectivity to wired and wireless devices. Equipped with a projector that enhances the learning experience with live instructors demonstrating coding, circuit designing, and systems integration. Infrastructures shall be installed to help resource-intensive tasks targeted for project-based learning on microcontrollers programming, IoT development, and robotics. It provides constant connectivity, smoothing the transmission of data during simulations and debugging of hardware, hence making it possible to have real-time testing and iteration by students. The ergonomic design of the layout ensures that students have enough space to work on hardware components and collaborate on complex projects with their peers. The lab is a highly workable space for hands-on learning and innovation, housing all tools relevant for advanced embedded system development.

3.0 Cable Description

Work Areas	Cable Type	Length (meters)
Ground Floor		
Hybrid Classroom	CAT6 Ethernet Cable	48
Video Conferencing Room	CAT6 Ethernet Cable	32
Student Lounge	CAT6 Ethernet Cable	17
Cafe	CAT6 Ethernet Cable	38
Inter-room connection	CAT6A Ethernet Cable	130
Total Length (Ground Floor)		265
First Floor		
General Purpose Lab 1	CAT6A Ethernet Cable	74
General Purpose Lab 2	CAT6A Ethernet Cable	74
Cisco Network Lab	CAT6A Ethernet Cable	46
Embedded Lab	CAT6A Ethernet Cable	59
Total Length (First Floor)		253
Fiber Optic Cable		150
Total Length (Overall)		668

4.0 Patch Cords and Switch Ports Description

Description	Quantity
Ground Floor	
Patch Cords	15
Switch Ports	24
First Floor	
Patch Cords	97
Switch Ports	4
Total (Patch Cords Overall)	112
Total (Switch Ports Overall)	28

5.0 Identifying Cable and Length

(a) CAT6 Ethernet Cable

For horizontal cabling, CAT6 Ethernet cables are used to connect devices within each work area. This cable type supports up to 10 Gigabit Ethernet speeds over distances up to 100 meters with a frequency of 250 MHz, ensuring high speed and reliable data transmission for all connected devices. The unshielded variant of CAT6 cables was chosen due to the minimal electromagnetic interference in the building environment. Its cost effective design and durable performance make it an excellent choice for network installations, providing reliability for at least 5 to 10 years.

Ground Floor Cable Lengths

- Hybrid Classroom: 48 meters
- Video Conferencing Room: 32 meters
- Student Lounge: 17 meters
- Cafe: 38 meters
- Inter-room Connection: 130 meters

Total Length: 265 meters

First Floor Cable Lengths

- General Purpose Lab 1: 74 meters
- General Purpose Lab 2: 74 meters
- Cisco Network Lab: 46 meters
- Embedded Lab: 59 meters

Total Length: 253 meters

(b) Fiber Optic Cable

Fiber optic cables are utilized for vertical backbone cabling, connecting network devices such as switches across floors. This cable type provides extremely high speed data transfer over long distances with minimal signal loss, making it ideal for inter floor connection. Its high bandwidth capacity ensures stable and rapid communication, critical for handling the large volume of data traffic in an academic environment,

- Total Fiber Optic Cable Length: 150 meters

(c) Wireless Network

Wireless Access Points (WAPs) are installed strategically in all work areas to provide seamless connectivity for mobile devices. These WAPs amplify Wi-Fi signals, ensuring strong and reliable wireless connectivity. The wireless network complements the wired infrastructure, offering flexibility and supporting a range of mobile devices without the need for additional cabling.

6.0 Meeting Minutes

DATE/TIME	22/1/2025 – 1720H		
LOCATION	Google Meet		
AGENDA	i) Discuss about Task 4 (mostly) and Task 5 ii) Decide who will to draw network diagram for each room iii) Decide who will explain the structure for all rooms iii) Divide task amongst the group members		
Meeting MC	Muhammad Afiq Danial bin Rozaidie		
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Muhammad Afiq Danish bin Mohd Hazni		1720H	-
Muhammad Syahmi Faris bin Rusli		1720H	-
Muhammad Afiq Danial bin Rozaidie		1720H	-
Dheshieghan A/L Saravana Moorthy		1720H	-
MINUTES			
NO.	ITEMS DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discuss the requirements of task 2 and task 3	Every member discusses the requirement to complete Task 4	Afiq Danish, Afiq Danial, Dhesh, Syahmi
2	Discuss about who in charge for ground floor	Afiq Danish and Syahmi will in charge to explain and draw for each room on ground floor	Afiq Danish, Syahmi
3	Discuss about who in charge for first floor	Afiq Danial and Dhesh will in charge to explain and draw for each room on first floor	Afiq Danial, Dhesh
4	Time to finish Task 4 to proceed to the Task 5	Every member must finish before or on due that has been set	Afiq Danish, Afiq Danial, Dhesh, Syahmi (Due 25/1/2025)

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4

TASK 4-MAKING THE CONNECTIONS – LAN and WAN (10 Marks)

In this task, you are putting all the infrastructure and devices into place. For example – where the switch is placed in the lab? How are these new rooms/networks connected to the main institution network? How much cabling do you need? How do you connect the different floors?

Once your group has chosen the internetworking devices you will need for your network, the devices must be interconnected. Your group needs to consider four physical areas when planning: work area, telecommunications room (distribution facility), backbone or vertical cabling, and distribution or horizontal cabling. The purpose of this discussion is for the groups to consider the physical connections that must take place for networking to be successful. Figures 1 and 2 gives you a basic idea of cabling structure.

You will also need to find out about what type of media (twisted pair, fibre optic, wireless) you need and its capabilities. Not all media are created equal and each has its own capabilities and limitations. Figures 3 and 4 shows you some examples.

Figure 1: Example of Wired LAN cabling structure

The diagram illustrates a wired LAN cabling structure. It shows a Server connected to a Patch Panel via a Patch cord. The Patch Panel is connected to an Ethernet Switch in a Closet via a Patch cord. The Ethernet Switch is connected to a Communications Outlet via a Patch cord. The Communications Outlet is connected to a Desktop PC via a Patch cord. UTP Cabling connects the Patch Panel and the Communications Outlet.

meet.google.com

MUHAMMAD SYAHMI FARIS BIN ...

DHESHIEGHA... Atiq Darul

MUHAMMAD ... MUHAMMAD ...